HALLIBURTON

Apr 23 2023 | Logged in as : Goytia, Renzo Nicolas NicolÃ¡s

Introduction

Baroid Core Homework 12

Assessment Feedback

Congratulations, you passed.

Total score: 73 out of 73, 100%

Question Feedback

Question

1 of 33

The study of fluid flow is called:

- ○ Plastic Viscosity
- ○ Gel strength
- ◉ Rheology
- ○ Yield Point

1 out of 1

Question

2 of 33

The fluid pumped through the bit nozzles while drilling is in transitional flow:

- ○ True
- ◉ False

1 out of 1

Question

3 of 33

What are the three basic types of flow regimes?

- ☐ Capillary flow
- ☑ Transitional flow

- [x] Laminar flow
- [x] Turbulent flow
- [ ] Stagnant flow

3 out of 3

Question

4 of 33

When the fluid moves in laminar flow, the motion is parallel to the walls of the channel and layers do not cross each other

- (•) True
- ( ) False

1 out of 1

Question

5 of 33

In the pits, the fluid moves about 6 to 300 rpm

- ( ) True
- (•) False

1 out of 1

Question

6 of 33

Viscosity can be understood as the relationship between a shear stress and a shear rate

- (•) True
- ( ) False

1 out of 1

Question

7 of 33

The main difference between a newtonian a non-newtonian fluid, is that newtonian fluids always exhibit a proportional relationship between shear stress and shear rate, and non-newtonian fluids also exhibit a non-proportional behavior

- (•) True
- ( ) False

1 out of 1

Question

8 of 33

Non-Newtonian fluids react in different ways under shear. For shear thinning fluids (also known as pseudoplastic) viscosity is time dependant

- ○ True
- ◉ False

1 out of 1

Question

9 of 33

When the three most widely used rheological models are compared, which one provides the most accurate estimations of pressure/ECD?

- ○ Bingham model
- ○ Power Law model
- ◉ Herschel-Bulkley model

1 out of 1

Question

10 of 33

The Bingham plastic rheological model usually over predicts the pressures of the system

- ◉ True
- ○ False

1 out of 1

Question

11 of 33

The Power law rheological model usually over predicts the pressures of the system

- ○ True
- ◉ False

1 out of 1

Question

12 of 33

From the following list, select all the different parameters that have an effect on the distribution of hydraulic energy on a drilling system

- [ ] Rate of fluid loss of the drilling fluid
- [x] Wellbore geometry
- [ ] Mineralogy of the rock
- [ ] Bottom hole temperature
- [ ] Hardness of the rock
- [x] Rig pumps
- [x] Bit selection and design
- [x] Eccentricity of the pipe
- [x] Rheological properties of the drilling fluid

5 out of 5

Question

13 of 33

Eccentricity is a dimensionless term that refers to the position of a pipe inside another pipe. In the oil field it usually refers to the position of the drillpipe in an annulus.

- (•) True
- ( ) False

1 out of 1

Question

14 of 33

The Fluid Jet Velocity is the average velocity of a drilling fluid passing through a bit's jet nozzles

1 out of 1

Question

15 of 33

The Bit Hydraulic Horsepower is the Bit hydraulic energy is the energy needed to counteract frictional energy (loss) at the bit.

1 out of 1

Question

16 of 33

What is the tubular pressure loss 1 (inside drill pipe)?

Tubular Pressure Loss 1 in psi = 2124

3 out of 3

Question

17 of 33

What is the tubular pressure loss 2 (inside drill collar)?

Tubular pressure loss 2 in psi = 665

3 out of 3

Question

18 of 33

What is the annular 1 pressure loss? (OH-DP)

Pressure loss annular 1 in psi = 62

3 out of 3

Question

19 of 33

What is the annular 2 pressure loss? (Casing-DP)

Pressure loss annular 2 in psi = 30

3 out of 3

Question

20 of 33

What is the annular 3 pressure loss? (Casing-DC)

Pressure loss annular 3 in psi = 8

3 out of 3

Question

21 of 33

What is the jet velocity in ft/s?

Jet velocity in ft/s = 150.94

3 out of 3

Question

22 of 33

What is the jet impact force?

Jet impact force in lb = 574.74

3 out of 3

Question

23 of 33

What is the bit pressure loss?

Bit pressure loss in psi= (2 decimals) 248.17

3 out of 3

Question

24 of 33

What is the bit hydraulic horsepower?

Bit hydraulic horsepower = 87.31

3 out of 3

Question

25 of 33

What is the bit horsepower per in^2?

Bit HPP/in^2= 0.74

3 out of 3

Question

26 of 33

What is the total system pressure loss?

Total pressure loss in psi = 3252

3 out of 3

Question

27 of 33

What is the ECD at the drill bit?

ECD_{bit} in ppg= 12.4

3 out of 3

Question

28 of 33

What is the ECD at the last casing shoe?

ECD_{shoe} in ppg = 12.39

3 out of 3

Question

29 of 33

What is the annular volume of the system in barrels?

Annular volume in bbl = 1222.69

3 out of 3

Question

30 of 33

What is the drill string volume?

Drill string volume in bbl = 151.47

3 out of 3

Question

31 of 33

What is the hole volume?

Hole volume in bbl = 1374.16

3 out of 3

Question

32 of 33

What is the total circulating volume?

Total circulating volume = 2574.16

3 out of 3

Question

33 of 33

What are the suction to suction strokes and time in minutes?

Strokes = 21542 Time in min= 180

1 out of 1